REFINING AN FPPF COVERING OF AN AFFINE SCHEME BY A STANDARD FPPF COVERING

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [2186, 2195] and proof(s) [[2197, 2201]]; this spec covers the complete environment instead (statement 2186-2195, proof span(s) [(2197, 2201)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Editorial note. The key mathematical input is cited; the assembly is compressed to one sentence in the source but the ingredients are named.

Source reference. Original reference 01UA: sources/stacks-chow/tags.txt, source lines 2403–2403. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let T be an affine scheme. Let $\{T_i \to T\}_{i \in I}$ be an fppf covering of T. Then there exists an fppf covering $\{U_j \to T\}_{j=1,\ldots,m}$ which is a refinement of $\{T_i \to T\}_{i \in I}$ such that each U_j is an affine scheme. Moreover, we may choose each U_j to be open affine in one of the T_i.*

Proof. This follows directly from the definitions using that a morphism which is flat and locally of finite presentation is open, see Morphisms, Lemma 01UA. □

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